Governing Monetary Flows: Informational Constraints and the Politics of Central Bank Digital Currencies

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Abstract

Why do states pursue central bank digital currencies (CBDCs), and why do their projects advance at different speeds? Accounts emphasizing monetary sovereignty, geoeconomic competition, and payment infrastructure functionality have primarily addressed the desirability and political significance of CBDCs in an ad hoc fashion. This article instead systematically traces variation in CBDC development to informational constraints in monetary governance. Because payments run through ledger updates distributed across institutions, an *informational constraint* arises when structural fragmentation or strategic leverage, domestic or international, limits an authority's effective reach into the records and channels through which money moves. These constraints generate pressures for institutional adaptation but operate differently across stages of CBDC development. Event-history models of 206 jurisdictions from 2008 to 2025 associate domestic payment fragmentation—a form of structural informational constraint—with CBDC investigation, and international financial sanctions—a form of strategic informational constraint—with advancement to a pilot or issuance. Official records connect these constraints to CBDC design. The findings suggest that weaponized interdependence can turn prior experimentation into strategic commitment by increasing the value of institutional alternatives.

1 Introduction

Over the past decade, central banks have raced to develop digital currencies. By 2025, 165 monetary authorities had initiated formal CBDC investigations, and 80 had advanced to a documented pilot or issuance. At least 16 state-led cross-border Digital Currency Payment Networks (DCPNs) were under development, with Europe, Hong Kong, and Singapore emerging as hubs of international collaboration.[1]

Despite the scale suggested by these aggregate figures, monetary authorities have followed markedly different trajectories in developing CBDCs. The Bahamas moved from an initial pilot to a nationwide launch within ten months[2]. Central banks in Canada and Denmark, while among the earliest to explore CBDCs, have deferred implementation after concluding that a retail CBDC currently offers insufficient added value or is not a near-term policy priority.[3]. Meanwhile, the United States has increasingly distanced itself from CBDC

[1]The author independently compiled and coded the data using central-bank reports, government documents, press releases, and the CBDC trackers maintained by the Human Rights Foundation and the Atlantic Council.

[2]Central Bank of The Bahamas 2019; Central Bank of The Bahamas 2020.

[3]Macklem 2024; Danmarks Nationalbank 2017.

development. The Federal Reserve began CBDC research and policy review in 2022 under the Biden administration. However, subsequent political opposition stopped further progress[4]. Policymakers have instead placed greater emphasis on private dollar-denominated stablecoins, presenting them as instruments to reinforce the international role of the dollar[5]. These divergences raise the article's central questions: Why do states pursue central bank digital currencies (CBDCs), and why do their projects advance at such different speeds?

Three groups of literature offer partial answers. Sovereignty-centered accounts read CBDCs as attempts to preserve public money against private digital currencies, foreign currencies, and foreign CBDCs[6]. Geoeconomic accounts place the adoption of CBDC within competition over international currencies and the strategic importance of infrastructures that transport cross-border payments[7]. Functional accounts emphasize payment efficiency, resilience, inclusion, and continued access to central-bank money as cash use declines[8].

These explanations, however, have mostly addressed CBDCs' desirability and political salience in an ad hoc fashion. They are also less explicit about why similar concerns produce different development trajectories or which considerations matter most at different stages, since central banks commonly articulate several objectives at once[9]. This article makes two analytical moves. First, it systematically traces variation in CBDC development by highlighting the underappreciated informational constraint in monetary governance. Second, it shifts attention from the objectives that authorities pursue to the institutional conditions under which those objectives become actionable. After all, existing accounts depend on an often implicit assumption: monetary authorities must be able to reach and use the information through which monetary activity is conducted and governed. Variation in that condition helps explain why similar objectives generate different institutional adaptation.

Money moves through information. Most contemporary payments are executed through updates to records maintained across financial institutions. Governing money requires *reach* into those records: the ability to obtain, interpret, and act on them. Reach is not conferred by legal authority or guaranteed by administrative resources. It depends on an authority's position relative to the records and channels through which its monetary system operates. A state may control currency issuance while relevant records remain within private platforms, arrive in incompatible formats, or reside in another jurisdiction. An *informational constraint* exists when institutional arrangements impede an authority's effective reach into the records and channels required to provide and govern monetary services.

Digitalization reshapes the arrangements on which that reach depends by lowering the costs of storing, transmitting, and verifying records[10]. It increases information volume and granularity and accelerates its production[11], while dispersing its custody across banks,

[4]Biden 2022; U.S. Congress, House 2023; Office of Congressman Tom Emmer 2025; Trump 2025.

[5]U.S. Congress, Senate 2025; U.S. Department of the Treasury 2025; Waller 2024.

[6]Marple 2021; Brooks 2021; Chia 2024; Chia and Helleiner 2024; Westermeier 2024; Fiebiger 2025.

[7]Peruffo, Cunha, and Ferrari Haines 2023; Deng 2024; Quaglia and Verdun 2025; Pforr, Pape, and Petry 2025; Bilotta 2025; Cong and Mayer 2025; Chey 2026.

[8]Auer, Frost, et al. 2022; Agur, Ari, and Dell'Ariccia 2022; Keister and Sanches 2023; Bank of Canada et al. 2020; Aurazo et al. 2024.

[9]European Central Bank 2020; Bank of England 2020; Board of Governors of the Federal Reserve System 2022.

[10]Goldfarb and Tucker 2019.

[11]Campbell-Verduyn, Goguen, and Porter 2017.

non-bank providers, wallets, and platforms[12]. Digital currencies can also shift traceability from following instructions across intermediary institutions to recording updates within ledger-based infrastructures[13]. Digitalization thus expands monetary information while redistributing control over the infrastructure. In such cases, monetary authorities may no longer enjoy the same degree of informational reach they once possessed.

Therefore, I argue that informational constraints can encourage monetary authorities to investigate and advance CBDCs as one means of restoring effective informational reach. These constraints vary along two dimensions. Their *source* is structural when they emerge from the evolution and organization of payment systems, and strategic when actors exploit their positions to condition or restrict access. Their *arena* is domestic when constraints operate among institutions within a jurisdiction, and international when they operate across borders. Crossing source and arena yields four types of informational constraints.

Specifically, *Domestic structural constraints* arise as payment records become fragmented across banks, non-bank providers, digital wallets, and platforms[14] under standards and reporting arrangements never designed to be read together[15]. *Domestic strategic constraints* arise when monetary activity is deliberately organized to operate outside supervised channels: permissionless systems record transfers in full while identifying no legal person and assigning no jurisdiction[16]. *International structural constraints* arise from routine dependence on foreign correspondent banks, messaging systems, and settlement institutions, which places parts of execution, record production, and compliance screening outside the authority's jurisdiction[17], a dependence that has been contracting for over a decade[18]. *International strategic constraints* arise when actors occupying central positions deliberately exploit this dependence to condition participation or interrupt access[19].

CBDC is one possible response to these constraints, not an automatic one. Authorities may instead tighten reporting, regulate platforms, build common clearing layers, modernize existing rails, or cooperate on cross-border standards. What makes a CBDC distinctive and attractive is that it can bring issuance, transfer, record-keeping, settlement, and authorized data access within a single publicly governed architecture, a unified ledger as the Bank of International Settlements advised, rather than repairing one link at a time[20].

The analytical framework produces stage-specific expectations. Structural constraints accumulate slowly and are diffuse. No single moment marks their arrival, and no single actor is responsible. They raise the value of *initiation*: a formal investigation into the new form of money, which is a comparatively cheap commitment that produces knowledge while preserving policy options. Strategic constraints are acute, dated, and attributable. They raise the urgency of *commitment* to a pilot or issuance of a workable CBDC, because the

[12]Feyen et al. 2021.

[13]Westermeier 2023.

[14]Langley and Leyshon 2021; Westermeier 2020.

[15]Bank of England 2021; European Central Bank 2024; Consultative Group on Innovation and the Digital Economy 2022.

[16]International Monetary Fund 2021; Financial Action Task Force 2021.

[17]Brandl and Dieterich 2023; Goede and Westermeier 2022.

[18]Rice, Peter, and Boar 2020; Committee on Payments and Market Infrastructures 2020.

[19]Farrell and A. L. Newman 2019; McDowell 2023.

[20]Consultative Group on Innovation and the Digital Economy 2023; BIS Innovation Hub and Bank of England 2023.

cost of the existing arrangement has become immediate and politically legible. But urgency is not sufficient. Commitment also requires legal authority, technical capacity, and agreement among public and private institutions. The conditions that place a CBDC on the agenda need not be the conditions that carry it toward operation.

I test this argument with a jurisdiction-year panel covering 206 monetary authorities from 2008 to 2025. Transition-specific event-history models estimate the initiation of a formal CBDC investigation and, among jurisdictions that have initiated, commitment to a pilot or issuance of such a currency. The models examine account-based finance, cryptocurrency restrictions, correspondent-banking contraction, prior Financial Action Task Force (FATF) exposure, active U.S./EU financial sanctions, and implementation capacity. Official decision records provide complementary evidence of how authorities diagnosed the problem, weighed alternatives, and translated that diagnosis into design.

The results support a stage-specific explanation of CBDC development under informational constraints. Across the models, the clearest pattern is that domestic fragmentation of payment systems helps place CBDCs on the policy agenda, whereas international financial pressure influences whether an existing project advances. Specifically, jurisdictions with higher levels of account ownership and digital-payment use tend to begin CBDC investigations earlier, but neither condition predicts advancement to a pilot or issuance. Active U.S./EU financial sanctions, by contrast, are not systematically associated with initiation but are associated with faster commitment among jurisdictions already investigating a CBDC. The relationship is concentrated in sanctions targeting financial institutions and payment channels rather than sanctions exposure generally, consistent with the proposed strategic informational constraint. This pattern emphasizes the acute informational constraint created by the weaponization of payment-network interdependence. Correspondent-banking contraction and prior FATF monitoring show no consistent relationship with either stage, while the findings for cryptocurrency restrictions vary across model specifications. Official decision records corroborate this interpretation: authorities explicitly connect fragmented payment systems and dependence on private or foreign infrastructure to CBDC evaluation and design. Taken together, the evidence suggests that financial pressure does not create an institutional alternative from scratch; it raises the strategic value of one already under development.

The article makes two contributions to the study of CBDCs and digital monetary governance. Theoretically, it systematizes the political logic of monetary governance by identifying a shared assumption of sovereignty-centered, geoeconomic, and functional accounts: all monetary authorities seek to maintain their effective influence over the systems they govern. The concept of *informational constraint* makes that premise explicit and connects domestic payment fragmentation, private platform power, and international infrastructural dependence within a common theoretical framework. Empirically, the article treats CBDC development as two sequential transitions rather than a single adoption decision. Combining a panel of 206 jurisdictions with official decision records reveals that the conditions associated with investigation differ from those associated with advancement to a pilot or issuance.

The findings also carry a broader implication for research on weaponized interdependence and institutional adaptation. In the CBDC setting, international financial pressure does not systematically place an institutional alternative on a state's agenda; it becomes consequential after prior investigation has made that alternative available. Coercive pressure may therefore accelerate existing institutional options but not create them.

The following sections review the three groups of literature and identify their shared premise, develop the theory and hypotheses, present the research design, report the quantitative and documentary evidence, and consider the implications for digital monetary governance.

2 Three Accounts of CBDC Development and Their Shared Premise

2.1 Sovereignty: Authority Without Reach

The first strand of literature situates digital currency within the relationship between money and political authority. State-centered monetary theory views money as rooted in organized relations of authority and credit rather than as a neutral medium of exchange[21]. Domestic monetary systems are inherently hierarchical in which central-bank liabilities provide the ultimate payment for settlement while commercial banks issue subordinate and conversable claims to state money[22]. That hierarchy was never purely public. Central banks have relied on private institutions to create credit, process payments, and transmit policy[23].

Digitalization alters these public-private relations. Permissionless cryptocurrencies attempt to validate transfers without a monetary authority; stablecoins combine privately governed payment arrangements with claims denominated in sovereign currencies[24]. Facebook's Libra proposal made the stakes vivid by suggesting an incumbent platform could technically scale a private payment instrument almost immediately[25]. More broadly, Fintech and Big Tech firms position their wallets and interfaces between users and banks, gaining control over payment initiation, routing, technical standards, and transaction data even when banks continue to provide accounts and final settlement[26]. Accordingly, CBDCs are attempts to preserve a central role for public money by recentering the payment activities around the central authority[27]. A retail CBDC could give the public direct access to central-bank liabilities while providing a public settlement anchor for private payment services. By contrast, a wholesale CBDC, likely restricted to eligible financial institutions, could reshape settlement in interbank and financial markets.

However, the monetary-sovereignty explanation identifies a policy objective rather than an achieved outcome, offering only an ad hoc account of the desirability and political salience of CBDC adoption. Issuance creates a public digital claim, but effective sovereignty also depends on legal authority, institutional design, intermediary participation, interoperability, and sustained use, none of which issuance alone guarantees. Early experience illustrates this distinction. Nigeria's eNaira remained operational after initial difficulties, but household adoption and merchant participation were limited[28]. China's e-CNY remained marginal

[21] Knapp 1924; Goodhart 1998; Helleiner 2003; Ingham 2004.
[22] Zimmermann 2013; Hockett and Omarova 2017.
[23] Strange 1988; Strange 1996; Braun 2020; Bernards and Campbell-Verduyn 2019.
[24] Chey 2022; Morgan 2023.
[25] Prasad 2021; Chia and Helleiner 2024.
[26] Bernards and Campbell-Verduyn 2019; Westermeier 2020.
[27] Chia and Helleiner 2024; Westermeier 2024; Fiebiger 2025; Cipollone 2024.
[28] Ree 2023.

relative to cash and established mobile-payment platforms, while the Eastern Caribbean's DCash pilot suffered from weak merchant integration, limited public awareness, and an extended technological outage. The Sand Dollar and JAM-DEX encountered similar problems with merchant onboarding and integration into existing financial institutions[29].

The pattern is consistent: formal monetary authority and operational reach are different things, and a state can hold the first while lacking the second. Sovereignty accounts explain why CBDCs become politically salient and desirable. They do not explain when salience becomes investigation, or when investigation becomes commitment.

2.2 Geoeconomics: Reach Seen from the Center

The second literature places CBDC development within a hierarchical international monetary system. The dollar's role rests on the depth of U.S. financial markets, its extensive use in trade and finance, and network effects around dollar-denominated settlement[30]. The hierarchy is infrastructural as well as monetary: cross-border transactions run through correspondent banks, messaging services, clearing houses, and settlement institutions concentrated in a few jurisdictions[31], while global finance depends on dollar-denominated credit created by both U.S. and non-U.S. institutions[32].

This concentration creates informational and coercive advantages: authorities controlling central nodes can monitor what passes through them and restrict access to them, the panopticon and chokepoint effects, respectively, in the theory of weaponized interdependence.[33]. U.S. authorities have used dollar clearing system and the position of American institutions to impose reporting requirements, freeze assets, and exclude targeted actors from access to the payment infrastructure and the dollar finance[34]. The growing use of financial sanctions has, in turn, spurred the development of alternatives, including China's Cross-Border Interbank Payment System, Russia's System for Transfer of Financial Messages, bilateral swap arrangements, and digital settlement platforms[35].

CBDC and cross-border digital-payment projects illustrate that digitalization has the potential to reorganize this hierarchy in different ways. Project mBridge establishes a shared ledger on which participating central and commercial banks can conduct wholesale CBDC payments and foreign exchange settlement directly, reducing reliance on correspondent banking chains and relocating settlement records and governance to the participating authorities[36]. Project Agorá follows a less disruptive approach. It combines tokenized central-bank reserves and commercial-bank deposits while preserving the two-tier banking system and the basic structure of correspondent banking. Its layered architecture shares information needed for coordination and settlement while leaving sensitive customer and compliance data with authorized institutions[37]. The comparison shows that digital payment infrastructures do

[29]Appendino et al. 2023; Koonprasert et al. 2024.
[30]Strange 1971; Cohen 1998; Helleiner 2008; Schwartz 2019.
[31]Brandl and Dieterich 2023; Petry 2021; Goede and Westermeier 2022.
[32]McCauley, McGuire, and Sushko 2015; Braun, Krampf, and Murau 2021; Murau, Rini, and Haas 2020.
[33]Farrell and A. L. Newman 2019.
[34]Cohen 2018; Goede 2021; Farrell and A. Newman 2023.
[35]McDowell 2023; Goede and Westermeier 2022; Pforr, Pape, and Petry 2025.
[36]Bank for International Settlements 2024a; Chu, Fang, and Lai 2025.
[37]Bank for International Settlements 2026.

not challenge the existing monetary order uniformly. Some create new shared settlement arrangements, while others modernize established banking relationships under a centralized digital framework. In both cases, their design shapes participation, information visibility, and the locus of regulatory authority.

These architectural differences are reflected in studies of individual CBDC projects. Research on the digital euro links the project to Europe's dependence on non-European payment providers and its pursuit of strategic autonomy[38]. Studies of the digital renminbi and digital ruble examine currency internationalization, state capacity, and efforts to reduce dependence on foreign infrastructure. They also document emerging relationships among central banks, commercial banks, and technology firms[39]. Collectively, this research shows that digital currencies are embedded in distinct political economies rather than constituting a uniform challenge to the existing international monetary system.

However, digital infrastructure alone does not confer monetary autonomy. Cross-border use continues to depend on convertibility, foreign-exchange liquidity, legal compatibility, common compliance arrangements, and the participation of foreign counterparties[40]. Whether any given project overcomes these constraints requires further investigation and falls outside the transitions examined here. The pattern of pursuit is nonetheless informative. Few of these projects could replace the incumbent infrastructures in the near term, but the number of jurisdictions examining alternatives has continued to grow, an indication that dissatisfaction with existing arrangements and the potential of the new technology are widespread. The literature primarily examines the power enjoyed by states that control the infrastructure of cross-border payments. Most jurisdictions pursuing CBDCs, however, occupy the opposite position: they depend on infrastructure controlled by others. For them, the central issue is not how control over payment networks generates leverage, but whether new digital infrastructure can extend their infrastructural reach, the capacity to transmit payment instructions, exchange compliance information, and complete settlement through reliable channels.

2.3 Function: Reach as a Precondition for Service

A third strand of literature focuses on the functional objectives of CBDCs: lowering payment costs and delays, strengthening resilience, expanding access for underserved populations, and preserving public access to central-bank money as cash use declines[41]. Evidence from mobile money shows that digital payment networks can reduce transaction costs, facilitate risk sharing, and reach users underserved by physical banking infrastructure[42]. Fintech and Big Tech firms have expanded these possibilities by integrating payments with mobile communications, e-commerce, lending, and other services[43].

These functional gains may nevertheless be accompanied by new coordination and gover-

[38] Westermeier 2024; Quaglia and Verdun 2025.

[39] Peruffo, Cunha, and Ferrari Haines 2023; Deng 2024; Ehlke, Salzer, and Westermeier 2025.

[40] Auer, Haene, and Holden 2021; Committee on Payments and Market Infrastructures et al. 2021; Ferrari Minesso, Mehl, and Stracca 2022.

[41] Auer, Frost, et al. 2022; Bank of Canada et al. 2020; Boar and Wehrli 2021; Di Iorio, Kosse, and Mattei 2024.

[42] Jack and Suri 2014; Suri and Jack 2016; Aron 2018.

[43] James and Quaglia 2024; Bernards and Campbell-Verduyn 2019.

nance problems. As payment functions become distributed across banks, payment providers, digital wallets, and platforms, different institutions may control customer interfaces, authenticate users, maintain accounts, route payment instructions, and provide final settlement. Platform operators acquire informational and technical advantages while remaining dependent on public and banking infrastructure.

Functional need must be distinguished from implementation capacity. Income, digital connectivity, bureaucratic quality, and financial development may facilitate CBDC development without generating a political impetus for adoption. Conversely, authorities may recognize deficiencies in existing payment systems but lack the resources or coordination required to build a reliable alternative. The IMF therefore treats CBDC development as a sequence of jurisdiction-specific decisions under uncertainty, while BIS research finds that fast-payment systems may either complement or substitute for retail CBDCs, depending on the existing payment ecosystem[44].

2.4 Information: the Shared Premise

Sovereignty-centered, geoeconomic, and functional accounts emphasize different features of monetary infrastructure but share a common assumption: that the institutions responsible for money can act upon it. Sovereignty accounts assume that authority translates into control; geoeconomic accounts, that network position translates into capability; and functional accounts, that appropriate design enables the system to deliver the intended service. Across these perspectives, the operative variable is whether the relevant records and channels lie within the reach of the institutions that need them.

Existing CBDC research treats information primarily as an output that a CBDC might provide, including greater transaction visibility, stronger state capacity, and a redistribution of informational control. Research on state legibility likewise examines what becomes visible to authorities and rightly emphasizes that records must be standardized, attributed, and connected before they can be used[45]. Visibility, however, is only one dimension of informational capacity. An authority may observe a transaction perfectly but remain unable to route or reconcile it, or to act on it. Treating information as an operational input rather than merely an observational output shifts the analytical focus from what states can see to what they can do. The next section develops the concept of informational constraints and clarifies their relationship to CBDC initiation and commitment.

3 Theory: Reach, Constraint, and CBDC Development

3.1 Money Moves as Information

The informational character of money predates digitalization. Money coordinates exchange by representing socially recognized claims and obligations[46]. In Giddens's formulation, "money proper" takes "the form of pure information lodged as figures in a computer printout" and

[44]Soderberg et al. 2023; Aurazo et al. 2024.

[45]Scott 1998; Fourcade and Gordon 2020; Goede 2018.

[46]Kocherlakota 1998; Simmel 2004.

allows economic relations to extend across time and space[47]. Most contemporary money is bank deposits, and most payments alter records on interconnected balance sheets rather than transferring any object[48]. Monetary infrastructures are therefore systems of record-keeping as much as systems of exchange: they represent claims, authenticate instructions, reconcile accounts, and settle obligations[49]. As Dodd observes, monetary networks are not containers of information but "networks of information"[50].

The consequence is easy to state and rarely followed through. If a payment is a record update, then the ability to perform and govern payments is the ability to act on records. Whoever holds the records, transmits them, defines their format, and controls who may act on them holds a share of monetary capability, whatever the law says about who issues the currency.

Digitalization creates a paradox of informational abundance and institutional dependence. It generates monetary records of unprecedented volume, granularity, speed, and linkability. Platforms can combine payment histories with data from commerce, communications, and lending, turning transactions into broader profiles of economic activity and reinforcing the data–network–activities loop identified by the BIS[51]. However, the proliferation of records does not make them equally usable by all institutions. Their operational value depends on access, standardization, interoperability, and jurisdiction. Data may be abundant while remaining fragmented across providers, encoded in incompatible formats, held under proprietary control, or located beyond an authority's legal reach.

This uneven distribution of informational capacity is a political-economic outcome rather than a neutral consequence of technological change. Monetary technologies emerge in response to the limitations of inherited arrangements, but their design also reflects and reshapes existing power relations[52]. By determining who can create, transmit, access, and act on monetary records, payment infrastructures connect Strange's credit and knowledge structures and redistribute the capacity to establish the conditions under which monetary claims circulate and settle[53]. The network constitutes a form of *knowledge regulation*: authority over the creation, dissemination, access, and use of information[54].

Institutional design gives this redistribution concrete form. The actors occupying central positions in payment systems shape both how transactions are executed and how their informational traces are produced and governed. Public authorities may depend on private institutions for the markets, instruments, and information through which monetary governance operates[55]. Digital currencies deepen this infrastructural dimension by shifting traceability from following transaction flows" across intermediaries to recording ledger updates," thereby embedding traceability within the payment architecture itself[56]. Digitalization thus expands the production of monetary information while potentially relocating control

[47]Giddens 1990, pp. 24–25.
[48]McLeay, Radia, and Thomas 2014; Minsky 1986; Ingham 2004; Murau and van 't Klooster 2023.
[49]Ali et al. 2014; Auer, Monnet, and Shin 2025.
[50]Dodd 1994, p. 26.
[51]Campbell-Verduyn, Goguen, and Porter 2017; Bank for International Settlements 2019.
[52]Marple 2021, pp. 439–441.
[53]Strange 1988; Strange 1990.
[54]Haggart and Germain 2026, p. 339.
[55]Braun 2020.
[56]Westermeier 2023, pp. 630–632.

over that information to banks, platforms, technical providers, or foreign infrastructures.

Research on state legibility captures part of this problem. Economic activity does not become governable simply because records exist; those records must be standardized, attributed, translated, and connected across institutional settings before authorized actors can use them. But payment governance requires more than visibility. Instructions must reach the appropriate institutions, records must be reconciled across systems, and authorized actors must be able to use them to complete monetary operations. When an institution lacks these capacities, it faces what the next section defines as an *informational constraint*.

3.2 Defining the Constraint

An *informational constraint* exists when institutional arrangements impede an authority's effective reach into the records and channels required to provide and govern monetary services.

Effective reach comprises four capabilities: ensuring that relevant records are produced, transmitting them to the institutions that need them, interpreting them upon arrival, and acting on them. In account-based monetary systems, payments are executed through coordinated record updates rather than the transfer of physical objects. Informational capabilities are therefore constitutive of monetary functions rather than merely adjacent to them. Because these capabilities form an operational sequence, a failure at any point can prevent an authority from performing the monetary function that depends on them. Four distinctions clarify the scope of the concept, while two additional features guide its empirical application.

First, effective reach is operational rather than merely observational. Research on state legibility examines how standardization, attribution, and connection make activities readable to the state, while research on financial surveillance considers how monitoring makes transactions visible to authorities. Effective reach encompasses these capacities but goes further by asking whether information can be used to authenticate instructions, transmit payments, and complete settlement.

Second, effective reach is distinct from, but closely related to, monetary sovereignty. Formal monetary sovereignty refers to the legal authority to issue money, define the unit of account, and establish monetary rules[57]. Murau and van 't Klooster reject this Westphalian conception in favor of *effective monetary sovereignty*: the capacity of states to govern different segments of a hybrid credit-money system, including money issued by regulated banks and by unregulated offshore institutions beyond the state's jurisdiction[58]. Effective reach specifies one condition of this capacity. To govern a segment of the monetary system, an authority must be able to act on the records and channels through which payments are executed. It may lack that ability even when its formal monetary authority remains uncontested.

Third, effective reach is distinct from state capacity. State capacity concerns the resources and organizational competence required to formulate and implement a response. An authority may recognize an informational constraint but lack the means to remedy it. The effective reach is not territorial: it concerns position within arrangements that cross boundaries of ownership and jurisdiction, and can be lost without any change in a state's ability to govern its own territory.

[57]Knapp 1924; Goodhart 1998; Zimmermann 2013.

[58]Murau and van 't Klooster 2023.

Fourth, effective reach is distinct from network dependence and varies by degree. Network dependence concerns the benefits and vulnerabilities created by relying on connections controlled by others[59]. Such dependence becomes an informational constraint only when it impedes one of the four capabilities identified above. Its severity also matters. Informational frictions may persist without motivating institutional change. China illustrates how such frictions can prompt a response. Non-bank payment providers maintained bilateral connections with individual commercial banks, leaving routing and reporting fragmented even as Alipay and WeChat Pay accumulated detailed transaction records. From 2017, the People's Bank of China required relevant online payments to clear through NetsUnion or China UnionPay, establishing a common layer for routing, standardization, and reporting[60]. NetsUnion is not a CBDC, but it shows that abundant records do not themselves provide effective reach. Detailed data existed in private hands, but the common routing and reporting layer required by the authority did not.

Two features govern the concept's empirical application. First, the analysis tests a narrower claim than the definition encompasses. Indicators available at the jurisdiction–year level capture conditions likely to impede informational reach rather than directly measuring the four informational constraints. The statistical models test the source-and-arena framework developed in the next section, in which each indicator has a specified referent. Second, an informational constraint is a feature of institutional arrangements, but it can motivate policy only when authorities recognize it. The analysis therefore combines statistical models, which assess whether the relevant conditions were present, with documentary evidence, which reveals whether authorities diagnosed those conditions as problems and invoked them when evaluating a CBDC.

3.3 Sources and Arenas

Two dimensions follow from the definition of informational constraints and together generate the framework

The first dimension concerns the *source* of the constraint. A constraint is **structural** when inherited arrangements fail to keep pace with changes in the scale or organization of monetary activity. It does not require deliberate obstruction by any particular actor. Instead, it arises when activity outgrows the institutions that support it. A constraint is **strategic** when an actor uses its position to withhold records, condition participation, redirect activity, or interrupt access. Structural dependence may create opportunities for strategic action, but the two sources remain analytically distinct: one arises from institutional drift, the other from purposive conduct.

The second dimension concerns the *arena* in which the constraint operates. A constraint is **domestic** when the relevant records and channels are located within domestic institutions, including banks, licensed payment providers, platforms, and privately governed networks. It is **international** when those records or channels are controlled by foreign correspondent banks, messaging systems, clearing institutions, or regulatory regimes.

Source and arena are independent dimensions. Any of the four informational capabilities

[59] Lake 2009; Farrell and A. L. Newman 2019.

[60] Bank for International Settlements 2019; People's Bank of China 2019; Chia and Wang 2026.

may be impeded by either a structural or strategic constraint operating in either the domestic or international arena. Source identifies how the constraint arises, while arena identifies where it operates; neither determines which capability it impedes. The two dimensions therefore intersect rather than compete. Table 1 presents the four resulting cases and identifies the capabilities each one typically impedes.

Table 1: Definitions of Informational Constraints by Source and Arena

Source	**Domestic arena**	**International arena**
Structural	**Domestic structural constraints** arise when inherited domestic arrangements fail to keep pace with changes in monetary activity, so that payment records are generated across banks, platforms, and payment providers under standards and reporting systems that were never built to be read together. No deliberate interference is required. *Links typically broken: production, interpretation.*	**International structural constraints** arise from routine reliance on foreign correspondent banks, messaging systems, clearing and settlement institutions, or legal regimes that place parts of payment execution and record production outside domestically usable channels. No deliberate interference is required. *Links typically broken: transmission, lawful use.*
Strategic	**Domestic strategic constraints** arise when monetary activity is deliberately organized to operate outside supervised channels, or when a private actor uses control over domestic platforms, interfaces, standards, or records to withhold information, condition participation, or redirect payment activity. *Links typically broken: production, lawful use.*	**International strategic constraints** arise when an actor controlling a central position in cross-border monetary networks uses that position to impose conditions on participation or to restrict access to messaging, liquidity, clearing, or settlement. *Links typically broken: transmission, lawful use.*

An informational constraint exists when institutional arrangements impede an authority's effective reach into the records and channels required to provide and govern monetary services. The rows identify the source of a constraint and the columns the arena in which it operates. The italicized tags name the links in the chain — production, transmission, interpretation, and lawful use — that each type most often degrades; any type may in principle degrade any link.

3.4 A CBDC as One Possible Response

An informational constraint creates a problem; it does not dictate a solution. Where records are fragmented across domestic providers, authorities may integrate reporting requirements, mandate a common clearing layer as the People's Bank of China did through NetsUnion, or modernize existing rails as the Eurosystem has done through T2 and TIPS, which settle

large-value and instant payments respectively in central-bank money[61]. Where cross-border dependence has narrowed, they may seek new correspondents, build regional platforms such as Afreximbank's PAPSS, which settles intra-African payments in local currencies rather than through a third currency, or interlink existing instant-payment systems, as Project Nexus does across five Asian jurisdictions[62]. Where access has been threatened or withdrawn, they may construct alternative messaging arrangements: Russia developed SPFS after 2014 and expanded it once selected banks were disconnected from SWIFT in 2022[63].

These responses are functionally adjacent: each reorganizes how payment instructions and records are transmitted, standardized, reconciled, and used, differing less in the informational problem addressed than in the institutional means of addressing it. But they operate within the existing architecture, connecting, substituting for, or modernizing channels while leaving intact the relation between central banks and the supervised intermediaries through which their money circulates. They also tend to repair one function at a time: for example, SPFS restored message transmission while leaving fund transfer dependent on correspondents. A CBDC operates at the level of the monetary claim itself. By introducing a digital central-bank liability, it can reallocate several connected payment and informational functions within an architecture whose core rules are publicly determined[64]. It is less a repair to the current system than a proposal for a different one.

That such a proposal is now under formal examination in 165 jurisdictions, with 80 having reached a pilot or issuance, indicates that this reconsideration has become general rather than exceptional. The question is therefore not whether authorities consider a CBDC, but when they judge it worth pursuing, and how far.

3.5 From Constraints to CBDC Engagement: Two Transitions

This article conceptualizes CBDC engagement as two transitions. *Initiation* occurs when a monetary authority begins a formal investigation. It is a relatively inexpensive and reversible step that produces knowledge about an unfamiliar instrument and preserves options without committing the authority to implementation. *Commitment* occurs when an initiated project advances to a pilot or issuance. This transition entails greater costs and institutional coordination, including the development of a legal and policy basis, a secure technological platform, interfaces with banks and payment providers, compliance procedures, and agreement over how responsibilities will be divided among participating institutions.

This distinction reflects the phased and iterative character of CBDC development. The IMF treats CBDC exploration as a sequence of decisions under uncertainty, with authorities able to continue learning, proceed, pause, or terminate a project at successive decision points[65]. Initiation and commitment therefore differ in cost and the extent of institutional coordination they require. Estimating the transitions separately permits this stage asymmetry to be observed rather than assumed; a single adoption outcome would combine the conditions

[61]European Central Bank 2025b; European Central Bank 2026.

[62]African Export-Import Bank 2021; Bank for International Settlements 2024b; Association of Southeast Asian Nations 2023.

[63]Cipriani, Goldberg, and La Spada 2023; Bank of Russia 2025.

[64]Consultative Group on Innovation and the Digital Economy 2023.

[65]Soderberg et al. 2023.

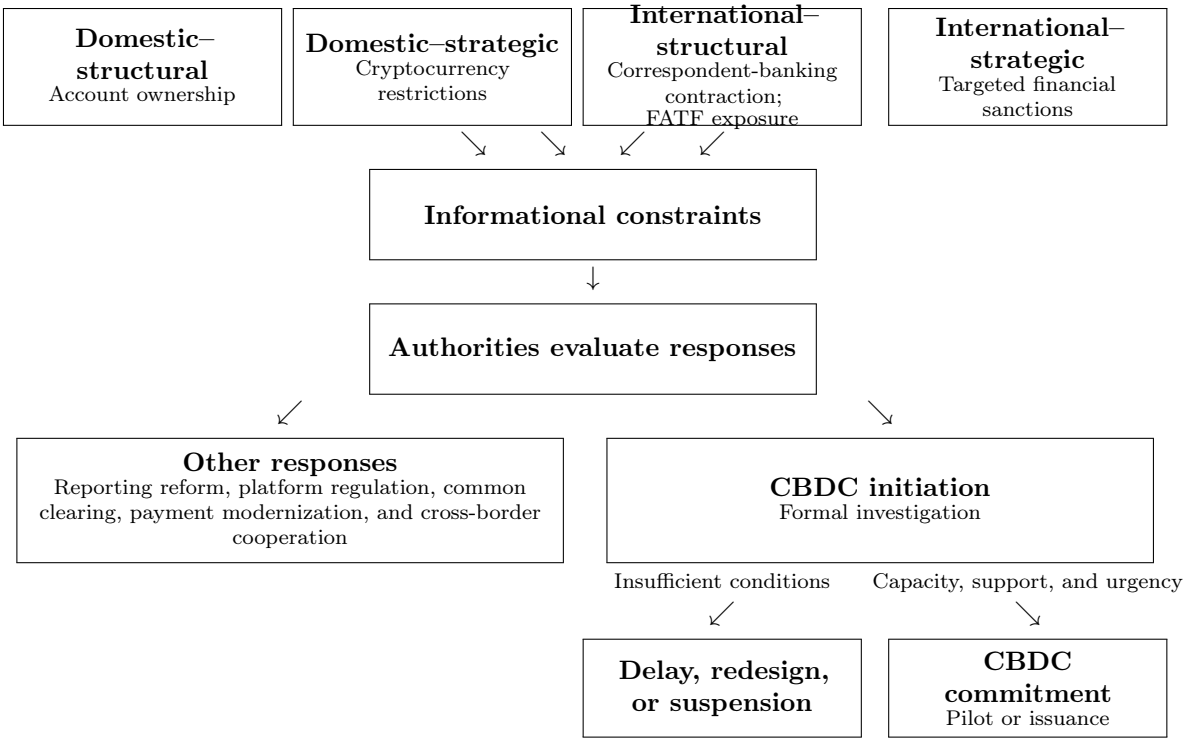


Figure 1: Informational Constraints and CBDC Development

Note: Informational constraints vary by source and arena. CBDCs are one possible institutional response. Progression from investigation to pilot or issuance remains conditional on implementation capacity, political support, and the urgency of the underlying constraint.

associated with agenda entry and implementation.

The following subsections derive stage-specific expectations from the four types of informational constraint. They identify which conditions should be most strongly associated with initiation and which should influence the transition from investigation to commitment. Figure 1 maps these types onto the development process, showing how they enter authorities' evaluation of alternative responses and may lead from formal CBDC investigation to pilot or issuance.

3.6 Hypotheses

Domestic structural. As monetary activity shifts from cash to accounts, a growing share of payments produces records across financial institutions and digital interfaces. Formal account ownership serves as a proxy for the scale of this account-based monetary system and, therefore, for a jurisdiction's potential exposure to informational fragmentation. As the system expands, the generation, custody, and processing of payment records may become dispersed across banks, nonbank providers, and platforms subject to different technical standards and regulatory obligations[66]. Central banks and international organizations describe the resulting frictions in terms of legacy data-collection systems, heterogeneous data models, incompatible

[66]Feyen et al. 2021; Langley and Leyshon 2021.

interfaces, and uncertain access rights[67].

Account ownership carries two components, and both point in the same direction for initiation. It measures the scale of the account-based system whose records must be coordinated, and it measures part of the institutional and user base that makes CBDC investigation practicable. The specification separates these from general readiness: logged real GDP holds constant the economic resources available for a technically demanding project, and the E-Government Development Index holds constant administrative and digital capacity, leaving account ownership to carry the financial-system-specific component. The appendix bears on the distinction more directly. If account ownership registered only a retail user base, it should not predict the initiation of wholesale projects, whose participants are financial institutions rather than households. In fact, its association is at least as strong for wholesale and cross-border investigations as for retail ones (Supplementary Appendix, Table S7). H1 therefore identifies a condition under which coordinating monetary information becomes more consequential, net of the general economic and administrative readiness with which it is correlated.

> *H1: Greater formal account ownership is associated with earlier initiation of a formal CBDC investigation.*

International structural. Cross-border monetary services rely on infrastructure distributed across multiple jurisdictions. Correspondent banks supply foreign-currency accounts, liquidity, compliance screening, clearing, and settlement, while messaging systems, such as the Society for Worldwide Interbank Financial Telecommunication (SWIFT), transmit the standardized instructions needed to coordinate these functions. A decline in correspondent relationships can interrupt payment channels, lengthen transaction chains, and redirect activity toward less transparent arrangements. The Committee on Payments and Market Infrastructures (CPMI) data show that the number of active correspondent relationships has declined globally even as cross-border message volumes have increased[68].

Financial Action Task Force (FATF) greylisting and blacklisting capture a related constraint: access to international payment systems may depend on compliance standards defined and enforced beyond a jurisdiction's borders. Listing publicizes deficiencies in a jurisdiction's financial-integrity framework, mobilizes market enforcement, raises compliance costs, and can restrict access to cross-border finance[69]. Longer listing periods are associated with greater losses of correspondent relationships[70]. Affected jurisdictions must therefore meet external informational and compliance requirements to maintain access to international payment channels.

> *H2a: Greater contraction in active correspondent-banking relationships is associated with earlier initiation of a formal CBDC investigation.*

[67] Bank of England 2021; European Central Bank 2024; Consultative Group on Innovation and the Digital Economy 2022; CGAP et al. 2024.

[68] Committee on Payments and Market Infrastructures 2020; Committee on Payments and Market Infrastructures 2023.

[69] Morse 2019.

[70] Rice, Peter, and Boar 2020.

H2b: Greater prior exposure to FATF monitoring or high-risk designation is associated with earlier initiation of a formal CBDC investigation.

Domestic strategic. Private crypto-assets can shift monetary activity beyond the supervised perimeter. Blockchains record on-chain data, such as wallet addresses, transaction amounts, and transfer times. These records do not identify the legal persons behind wallet addresses or locate them within particular jurisdictions. In short, they do not contain "know your customer" (KYC) information. Payers' identities can still be reconstructed using off-chain information. This information is dispersed across exchanges, custodians, wallet providers, and other intermediaries.[71].

In this case, I use government attitudes toward cryptocurrency to gauge state policy toward private cryptocurrencies. A restrictive policy is an observable response to digital monetary activity outside the supervised perimeter. Governments impose restrictions for varied reasons. These include financial stability, consumer protection, illicit finance, capital flight, currency substitution, and energy use[72]. Despite this variation, restrictions share an effort to limit or condition activity through privately governed channels[73]. They indicate that an authority views such activity as requiring intervention. I therefore expect authorities with restrictive policies to investigate CBDCs earlier and move more quickly toward a publicly authorized alternative. The analysis tests this expectation at both stages and across alternative specifications.

H3a: Prior adoption of a restrictive policy toward private cryptocurrencies is associated with earlier initiation of a formal CBDC investigation.

H3b: Among jurisdictions that have initiated a CBDC investigation, prior adoption of a restrictive cryptocurrency policy is associated with faster advancement to a pilot or issuance.

International strategic. Financial sanctions are the clearest case of an external actor deliberately conditioning access to monetary infrastructure. Control over correspondent accounts, custodial services, and messaging networks can be used to block transactions or exclude designated actors from international payments[74]. Sanctions enforcement also depends on information. Transaction monitoring and designation lists allow banks and public agencies to identify targeted actors, relationships, and assets. Financial intermediaries therefore become participants in sanctions enforcement[75]. Banks may respond by intensifying screening, terminating relationships, or avoiding transactions beyond what the law formally requires.

Financial sanctions most closely match the proposed mechanism when they operate through financial institutions, payment channels, or access to settlement. Their effects should be concentrated at the commitment stage rather than the initiation stage. CBDC investigation is relatively inexpensive and already widespread. A concrete disruption, however, can reveal

[71]International Monetary Fund 2021; Financial Action Task Force 2021.
[72]International Monetary Fund 2023; Ba and Şen 2024; Hebous and Vernon 2023.
[73]International Monetary Fund and Financial Stability Board 2023.
[74]Cipriani, Goldberg, and La Spada 2023.
[75]Amicelle 2011; Amicelle and Jacobsen 2016.

operational dependence, mobilize affected institutions, and increase the urgency of advancing an existing project. Comparing financial and non-financial sanctions provides a direct test of this mechanism. If the relationship is driven by disruption to monetary infrastructure rather than geopolitical pressure in general, financial sanctions should have a stronger association with commitment than nonfinancial sanctions.

> *H4: Among jurisdictions that have initiated a CBDC investigation, exposure to an active U.S. or EU financial-sanctions case is associated with faster advancement to a pilot or issuance.*

H4 does not imply that a CBDC can neutralize sanctions or that every financial-sanctions case directly targets payment infrastructure. The claim is narrower: pressure from authorities occupying central positions in the networks built around the dollar and the euro, the two most widely used international currencies, can raise the perceived costs of existing dependencies and strengthen incentives to invest in alternative monetary infrastructure.

3.7 Measurement Strategy

The empirical strategy translates each type of informational constraint into an observable, mechanism-specific indicator. Account ownership captures the scale of the account-based monetary system that an authority must coordinate. Correspondent-banking contraction and prior FATF exposure capture declining access to cross-border channels and stricter conditions of participation. Cryptocurrency restrictions identify official intervention in privately governed digital monetary activity. Active financial sanctions capture deliberate external pressure on access to monetary infrastructure. These indicators do not form a single index. Each represents a different combination of source and arena and generates a distinct stage-specific expectation.

Account ownership also reflects the institutional base that makes CBDC investigation practicable. The models therefore include GDP and e-government development as separate controls. These variables help distinguish the scale of account-based finance from general economic resources and administrative readiness. The sanctions mechanism likewise concerns access to financial and payment infrastructure rather than geopolitical pressure in general. The analysis therefore distinguishes financial from non-financial sanctions. If the proposed mechanism is correct, the association with commitment should be concentrated among financial-sanctions cases rather than sanctions of all types.

Estimating initiation and commitment as separate transitions allows stage asymmetry to be observed rather than assumed. A single adoption outcome would conflate the conditions associated with placing a CBDC on the policy agenda with those associated with advancing an existing project toward operation. The quantitative analysis tests these stage-specific relationships. Official decision records then assess whether monetary authorities diagnosed the relevant conditions and connected them to CBDC evaluation and design.

4 Research Design

4.1 Sample and Unit of Analysis

The analysis uses an annual panel of 206 monetary jurisdictions observed from 2008 through 2025, producing 3,708 jurisdiction-year observations before event-history censoring and missing-data restrictions. The sample contains all 193 United Nations member states and 13 additional jurisdictions for which monetary institutions and CBDC activity can be observed.[76] The panel begins in 2008, the year of the Bitcoin white paper was released, providing a common pre-investigation baseline before state-led CBDC activity became widespread.

The dependent variable separates CBDC development into two transitions: *initiation* and *commitment*. Initiation is the first year in which a monetary authority or government publicly begins a formal CBDC research or investigation process. [77] Commitment is the first year in which an initiated project advances to a publicly documented pilot or issuance. [78][79][80]

In the snapshot estimation used for the analysis, 165 jurisdictions initiate a CBDC investigation and 80 reach the commitment threshold by the end of 2025. The median interval from initiation to commitment among jurisdictions that progress is three years. Table 2 distinguishes these full-sample transition counts from the complete-case risk sets used to estimate the initiation and commitment models. [81]

Table 2: CBDC Transition Risk Sets, 2008–2025

Transition	Risk set	Events	Right-censored	Event share
Initiation	206	165	41	80.1%
Commitment	165	80	85	48.5%

[76]The additional jurisdictions are Anguilla, Aruba, Bermuda, the Cayman Islands, Curaçao, Greenland, Hong Kong, Kosovo, Macao, Montserrat, Palestine, Sint Maarten, and Taiwan.

[77]Qualifying evidence includes a central-bank report, a public consultation paper, the establishment of an official working group, the start of a feasibility project, or a comparably specific research program, provided that a public announcement states the beginning of a state-led digital currency program.

[78]For coding purposes, commitment occurs in the first year that a monetary authority officially announces the launch, formal approval, or scheduled commencement of a defined CBDC pilot, or announces formal issuance.

[79]CBDC histories were assembled from dated central-bank, finance-ministry, and government publications. The Atlantic Council and Human Rights Foundation trackers were used to identify candidate cases and documents (Atlantic Council GeoEconomics Center 2025; Human Rights Foundation 2025).

[80]When sources disagree, a dated primary source that clearly satisfies the relevant threshold takes precedence over trackers and secondary reports. The earliest reported year is coded only when the underlying evidence meets the substantive coding rule; otherwise, the earliest corroborated date is used. The online dataset provides the complete jurisdiction inventory and supporting sources.

[81]The outcomes record the first occurrence of each qualifying transition rather than a project's current status. Once a transition has occurred, a subsequent suspension or discontinuation does not erase that earlier event. Denmark concluded its e-krone investigation without proceeding to a pilot and is therefore coded as initiated but not committed. Ecuador later discontinued its operational Dinero Electrónico system, but the project remains coded as having crossed the commitment threshold. Because cancellations are rare, they are not modeled as a separate or competing transition.

4.2 Explanatory Variables

The empirical measures follow the structural–strategic and domestic–international distinctions developed in the Theory section.

Domestic structural conditions. The primary domestic structural measure is the share of adults aged 15 or older who report having an account in a bank or similar financial institution, drawn from the World Bank's Global Findex[82]. Account ownership captures the reach of account-based monetary activity and the institutional and user base through which a CBDC could operate. Extensive account ownership may improve public legibility while also increasing the volume and importance of records distributed across heterogeneous providers. The variable is lagged one year and expressed in ten-percentage-point increments.

To support the argument about domestic digitalization, an alternative specification replaces account ownership with the share of adults who made or received a digital payment, also drawn from the World Bank's Global Findex. This measure more directly captures the diffusion of data-generating payment activity, but it does not distinguish among platform payments, bank transfers, card payments, and mobile money.[83]

International structural conditions. Two measures describe a jurisdiction's structural position in cross-border finance. The first is the cumulative percentage loss of active correspondent-banking relationships relative to 2011. It is derived from the Committee on Payments and Market Infrastructures (CPMI), coded so that higher percentage values indicate a larger contraction, lagged one year, and expressed in ten percentage-point increments[84]. This measure captures the declining access to regulated cross-border channels. The source series ends in 2022, and its final observed value is carried forward through 2025. After applying the lag, no new source observation enters the measure after 2023.

The second measure, Financial Action Task Force (FATF) exposure, is a cumulative count of the years since 2008 in which a jurisdiction has appeared on the FATF's grey or black list, lagged one year and transformed using $\log(1 + x)$ to reduce the influence of long-running cases[85]. FATF exposure captures external scrutiny of a jurisdiction's financial-integrity and reporting framework. Listing can increase the due-diligence costs imposed by foreign institutions and decrease the willingness of foreign cross-border payment providers to engage with a listed jurisdiction.

Domestic strategic contestation. Domestic strategic contestation is measured by the adoption of a partial or comprehensive legal restriction on private cryptocurrency activity.

[82] Demirgüç-Kunt et al. 2022.

[83] The Global Findex reports account ownership in five survey rounds: 2011, 2014, 2017, 2021, and 2024. Digital-payment use is available in four rounds: 2014, 2017, 2021, and 2024. In the dataset, a jurisdiction's first observed value is carried forward until a later survey provides an update. Years before the first observation remain missing. Future observations are not used to fill earlier gaps. Account ownership and digital-payment use enter the models separately because they measure related but distinct aspects of financial activity, which may increase concern of multicollinearity if both are presented in one specification.

[84] Committee on Payments and Market Infrastructures 2023.

[85] Financial Action Task Force 2025.

Such restrictions indicate an effort to reassert public authority over monetary activity conducted through privately governed channels. Partial restrictions often target the interfaces between cryptocurrency markets and the regulated financial system, including banks, exchanges, custodians, and payment providers. They condition who may participate, which transactions may be processed, and what information intermediaries must provide. Comprehensive restrictions pursue the same objective through broader prohibitions. Both therefore provide observable evidence of contestation over access to and control of digital monetary activity.

The indicator remains coded as one after the first year in which a restriction is observed. It captures the decision to intervene rather than the government's specific motivation. The primary indicator is lagged by one year to avoid treating a restriction adopted in the same year as a CBDC transition as temporally prior to that transition.

International strategic contestation. International strategic contestation is measured using sanctions cases from Release 4 of the Global Sanctions Data Base (GSDB-R4)[86]. GSDB records sanctions as policy episodes rather than counts of designated persons or entities and identifies each episode's sender, target, duration, and type. The analysis retains cases imposed by the United States, the European Union, or the European Economic Community because their currencies and financial systems occupy central positions in cross-border finance.

Cases naming a targeted single state are assigned directly to that jurisdiction. For multi-state cases beginning in 2004 or later, each named constituent state is coded separately. Because GSDB-R4 ends in 2023, official sources are used to extend the dataset through 2025. Cases confirmed to remain in force are coded as active through 2025. Cases terminated in 2024 or 2025 are assigned the relevant end year. New U.S. and EU cases beginning during these years are collected from official Office of Foreign Assets Control (OFAC) records and Council of the European Union legal acts and announcements. They are coded using the same rules for senders, targets, duration, and sanctions type[87].

A case is active from its start year through its end year. The data are then aggregated to the jurisdiction–year level. The primary measure equals one when at least one U.S. or EU financial-sanctions case was active in the previous year. The broad measure includes all U.S. and EU sanctions cases. The decomposed model includes financial and non-financial cases as separate indicators. The categories are mutually exclusive at the case level, but both may be active in the same jurisdiction–year. All measures are lagged by one year.

Covariates. The models include the United Nations E-Government Development Index[88]. The index ranges from zero to one and combines online services, telecommunications infrastructure, and human capital. The United Nations reports it in biennial survey waves from 2008 through 2024. Each observed value is carried forward until the next survey wave. Years before a jurisdiction's first observation remain missing, and future values are not used to fill earlier years. The models also include two indicators from the World Bank's World

[86]Felbermayr et al. 2020; Yalcin et al. 2024.

[87]U.S. Department of the Treasury, Office of Foreign Assets Control 2026; Council of the European Union 2026.

[88]United Nations Department of Economic and Social Affairs 2024.

Development Indicators: GDP in constant 2015 U.S. dollars and annual consumer-price inflation[89]. E-government development captures broad administrative and digital capacity. Logged GDP captures the resources available for a technically demanding project. Inflation accounts for macroeconomic conditions that may affect the salience of monetary reform.

All time-varying covariates are lagged by one year. Account ownership is measured in ten-percentage-point increments. The E-Government Development Index is rescaled so that one unit represents a 0.10-point increase. Real GDP is logged and standardized. Inflation is winsorized at the first and ninety-ninth percentiles. Current-dollar GDP is used only to normalize the nominal BIS locational-statistics measures reported in the appendix.

Table 3 summarizes the variables, their operationalization, scale, and their role in the empirical analysis.

Table 3: Summary of Variables and Empirical Measures

Variable	Operationalization	Scale	Role
CBDC transitions			
Initiation	First year of a formal CBDC investigation	Annual first event	Dependent variable
Commitment	First year of a documented CBDC pilot or issuance	Annual first event after initiation	Dependent variable
Structural conditions			
Account ownership	Adults with a financial-institution or mobile-money account	$t-1$; per 10 p.p.	Domestic; H1
Digital-payment use	Adults who made or received a digital payment	$t-1$; per 10 p.p.	Alternative H1 measure
Correspondent-banking loss	Cumulative decline in active correspondent-banking relationships from 2011	$t-1$; per 10 p.p.	International; H2a
FATF exposure	Cumulative years under FATF enhanced monitoring or a call for action	Through $t-1$; $\log(1+x)$	International; H2b
Strategic contestation			
Cryptocurrency restriction	Partial or comprehensive legal restriction on private cryptocurrency activity	$t-1$; binary	Domestic; H3a–H3b
Active U.S./EU financial sanctions	At least one active U.S.- or EU-sender financial-sanctions case against the jurisdiction	$t-1$; binary	International; H4
Broad and nonfinancial sanctions	Any active U.S.- or EU-sender case; broad exposure is decomposed into financial and nonfinancial cases	$t-1$; binary	H4 comparison
Implementation capacity and covariates			
E-government capacity	United Nations E-Government Development Index	$t-1$; per 10 p.p.	Implementation capacity
GDP	Gross domestic product, logged	$t-1$; standardized	Economic capacity
Inflation	Annual consumer-price inflation	$t-1$; winsorized at 1st/99th percentiles	Macroeconomic covariate

4.3 Transition-Specific Event-History Models

Initiation and commitment are sequential stages of CBDC development. They are not repeated occurrences of the same event. Jurisdictions first move from no formal investigation to initiation. They may then move from initiation to commitment. Each transition has its own risk set, duration clock, baseline hazard, and covariate effects[90]. I estimate a separate

[89] World Bank 2026a; World Bank 2026b.

[90] Metzger and B. T. Jones 2016; Andersen and Keiding 2002.

Cox proportional-hazards model for each transition. Both models include time-varying covariates[91]. The data are organized into annual jurisdiction–year intervals.

For initiation, jurisdictions enter the risk set in 2008. They leave when they begin a formal CBDC investigation or are right-censored at the end of 2025. Duration is measured in years since 2008. For commitment, jurisdictions enter the risk set once they initiate an investigation. The duration clock begins in the initiation year. It ends at the first pilot or issuance, or at censoring in 2025. The commitment model therefore estimates how quickly a project advances beyond investigation. When initiation and commitment occur in the same year, commitment is coded as an event in the first interval.

For transition s and specification m, the conditional hazard is

$$h_{is}^{(m)}\left(\tau \mid \mathbf{X}_{ist}^{(m)}\right) = h_{0s}^{(m)}(\tau)\exp\left(\mathbf{X}_{ist}^{(m)\prime}\boldsymbol{\beta}_{s}^{(m)}\right), \qquad s \in \{\text{initiation}, \text{commitment}\}, \tag{1}$$

where τ denotes transition-specific duration, $h_{0s}^{(m)}(\tau)$ is the unspecified baseline hazard, and $\mathbf{X}_{ist}^{(m)}$ contains the predictors included in specification m. Coefficients and baseline hazards are estimated separately for each transition and specification. Ties are handled using the Efron approximation. Standard errors are robust and clustered by jurisdiction.

The first model sequence evaluates the two dimensions of the theoretical framework. Let A denote account ownership, R the cryptocurrency restriction, and $\mathbf{Q}$ the international structural measures. The latter include correspondent-banking loss and prior FATF exposure. Let $\mathbf{W}$ denote the capacity and macroeconomic covariates, and S active U.S./EU financial-sanctions exposure. The four predictor sets are

$$\begin{aligned} \mathcal{X}_{s}^{\text{Domestic}} &= \{A, R, \mathbf{W}\}, \\ \mathcal{X}_{s}^{\text{International}} &= \{\mathbf{Q}, S, \mathbf{W}\}, \\ \mathcal{X}_{s}^{\text{Structural}} &= \{A, \mathbf{Q}, \mathbf{W}\}, \\ \mathcal{X}_{s}^{\text{Strategic}} &= \{R, S, \mathbf{W}\}. \end{aligned} \tag{2}$$

These specifications show whether the results follow the domestic–international distinction, the structural–strategic distinction, or both.

A second model sequence holds the remaining predictors and estimation sample constant while varying the sanctions measure. It compares broad U.S./EU sanctions exposure with financial-sanctions exposure. It also estimates a decomposed model containing financial and nonfinancial exposure together. For sanctions measure or vector $S^{(k)}$, the predictor set is

$$\mathcal{X}_{s}^{(k)} = \{A, R, \mathbf{Q}, S^{(k)}, \mathbf{W}\}. \tag{3}$$

The primary specification uses active U.S./EU financial-sanctions exposure at $t-1$. The broad and financial indicators are estimated in separate models because financial cases are nested within the broad category. The decomposed model enters financial and nonfinancial exposure as separate indicators. The case categories are mutually exclusive, but both indicators may equal one in the same jurisdiction–year when separate cases are active. The model also tests whether their coefficients differ. Further alternatives distinguish U.S. from

[91] Box-Steffensmeier and B. S. Jones 2004; Metzger and B. T. Jones 2022.

EU senders. They also examine active-case counts, new case onsets, recent onsets, cumulative exposure, a manually audited infrastructural indicator, and the former OFAC institution-level series.

4.4 Missing Data and Covariate Coverage

Table S1 in the Supplementary Appendix reports covariate coverage in the full jurisdiction–year panel. Missingness is concentrated in the Global Findex account-ownership and correspondent-banking series. Supplementary Appendix Table S2 assesses whether this limited coverage affects the estimates for the remaining covariates. It omits each measure separately and then omits both together.

The primary Findex series carries each jurisdiction's most recent survey observation forward until a new observation becomes available. Years before the first observation remain missing. Future observations are not used to fill earlier years. The checked data corrects one source cell for the United Arab Emirates' 2024 account-ownership value. Recomputing the carried-forward and lagged series changes only one transformed model cell. The dependent variables are never imputed.

For other covariates, the checked panel supplements 205 missing historical GDP and inflation cells with IMF WEO observations. These observations are used only when the metadata identify a national statistics office or central bank as the underlying source. Their year must also be no later than the latest-actual marker. The panel estimates 90 bounded one- or two-year gaps in the supplementary bank-ownership and BIS locational series using adjacent-year interpolation. These estimates do not enter the preferred specification. No e-government survey-wave gaps are interpolated.

5 Results

The results reveal a stage-specific pattern. In general, structural constraints are more associated with the initiation of CBDC investigation, while strategic constraints are more correlated with the commitment to CBDC pilot and issuance. The conditions that place a CBDC on the policy agenda differ from those that carry an initiated project toward implementation. The initiation models contain 1,429 jurisdiction-years, 154 jurisdictions, and 132 initiations. The commitment models contain 618 post-initiation jurisdiction-years, 133 jurisdictions, and 64 commitments. All sanctions comparisons use the same transition-specific samples and one-year-lagged measures. The following layouts of the empirical results are presented in the structural–strategic and domestic-international comparisons.

The results are presented in three figures and two tables. The figures summarize the principal associations, while Tables 4 and 5 report the complete estimates. The first table compares the Domestic, International, Structural, and Strategic specifications. The second compares broad, financial, and decomposed sanctions measures. In both tables, Panel A reports initiation and Panel B reports commitment.

Table 4: Domestic, International, Structural, and Strategic Models of CBDC Development

	Domestic	International	Structural	Strategic
Panel A. Initiation of CBDC investigation				
Account ownership, 10 pp	1.221*** [1.094, 1.364]	—	1.210*** [1.082, 1.353]	—
Cryptocurrency restriction, $t-1$	1.487 [0.898, 2.463]	—	—	1.459* [0.931, 2.288]
Correspondent-banking loss, 10 pp	—	0.973 [0.888, 1.066]	0.971 [0.881, 1.071]	—
Prior FATF exposure, $\log(1+x)$	—	0.845 [0.680, 1.050]	0.855 [0.695, 1.052]	—
Active financial-sanctions case, $t-1$	—	0.806 [0.528, 1.230]	—	0.740 [0.481, 1.138]
E-government capacity, 10 pp	0.954 [0.810, 1.124]	1.132** [1.002, 1.278]	0.949 [0.794, 1.134]	1.151** [1.024, 1.293]
Inflation	1.007 [0.991, 1.022]	1.010 [0.994, 1.027]	1.012 [0.997, 1.026]	1.007 [0.990, 1.024]
Real GDP (constant 2015 USD), logged and standardized	1.410** [1.062, 1.872]	1.424** [1.058, 1.917]	1.431** [1.057, 1.937]	1.385** [1.038, 1.847]
Jurisdiction–years	1429	1429	1429	1429
Jurisdictions	154	154	154	154
Initiations	132	132	132	132
Panel B. Commitment to a pilot or issuance				
Account ownership, 10 pp	1.091 [0.886, 1.344]	—	1.000 [0.822, 1.217]	—
Cryptocurrency restriction, $t-1$	2.574** [1.092, 6.063]	—	—	1.862 [0.783, 4.426]
Correspondent-banking loss, 10 pp	—	1.113 [0.895, 1.383]	1.117 [0.893, 1.396]	—
Prior FATF exposure, $\log(1+x)$	—	1.040 [0.708, 1.527]	1.121 [0.758, 1.660]	—
Active financial-sanctions case, $t-1$	—	1.954** [1.125, 3.394]	—	1.796* [0.984, 3.279]
E-government capacity, 10 pp	1.276* [0.981, 1.658]	1.330*** [1.095, 1.614]	1.269* [0.964, 1.670]	1.436*** [1.189, 1.735]
Inflation	1.006 [0.989, 1.024]	0.997 [0.974, 1.021]	1.003 [0.981, 1.026]	1.002 [0.985, 1.020]
Real GDP (constant 2015 USD), logged and standardized	1.030 [0.718, 1.476]	1.367* [0.974, 1.917]	1.346* [0.946, 1.914]	1.100 [0.761, 1.590]
Jurisdiction–years	618	618	618	618
Jurisdictions	133	133	133	133
Commitments	64	64	64	64

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Hazard ratios; 95% confidence intervals in brackets.

Table 5: Broad, Financial, and Decomposed Sanctions Specifications

	Broad	Financial	Decomposed
Panel A. Initiation of CBDC investigation			
Account ownership, 10 pp	1.218*** [1.086, 1.365]	1.215*** [1.085, 1.362]	1.214*** [1.083, 1.360]
Cryptocurrency restriction, $t-1$	1.812** [1.078, 3.044]	1.837** [1.102, 3.062]	1.891** [1.120, 3.192]
Correspondent-banking loss, 10 pp	0.966 [0.876, 1.064]	0.968 [0.878, 1.068]	0.964 [0.877, 1.061]
Prior FATF exposure, $\log(1+x)$	0.807* [0.648, 1.005]	0.811* [0.653, 1.007]	0.808* [0.649, 1.006]
Broad active sanctions case, $t-1$	0.921 [0.621, 1.366]	—	—
Active financial-sanctions case, $t-1$	—	0.823 [0.540, 1.256]	0.798 [0.514, 1.239]
Active nonfinancial-sanctions case, $t-1$	—	—	1.180 [0.647, 2.153]
E-government capacity, 10 pp	0.949 [0.798, 1.130]	0.946 [0.797, 1.123]	0.956 [0.807, 1.133]
Inflation	1.011 [0.997, 1.026]	1.012 [0.997, 1.027]	1.013* [0.998, 1.028]
Real GDP (constant 2015 USD), logged and standardized	1.377** [1.027, 1.847]	1.377** [1.028, 1.844]	1.365** [1.018, 1.830]
p: financial = nonfinancial	—	—	0.359
Jurisdiction–years	1429	1429	1429
Jurisdictions	154	154	154
Initiations	132	132	132
Panel B. Commitment to a pilot or issuance			
Account ownership, 10 pp	1.091 [0.875, 1.360]	1.088 [0.873, 1.355]	1.111 [0.887, 1.390]
Cryptocurrency restriction, $t-1$	2.104 [0.757, 5.852]	2.036 [0.739, 5.605]	2.241 [0.790, 6.354]
Correspondent-banking loss, 10 pp	1.084 [0.871, 1.351]	1.083 [0.871, 1.348]	1.089 [0.872, 1.359]
Prior FATF exposure, $\log(1+x)$	0.937 [0.611, 1.439]	0.951 [0.620, 1.458]	0.960 [0.624, 1.477]
Broad active sanctions case, $t-1$	1.688* [0.926, 3.079]	—	—
Active financial-sanctions case, $t-1$	—	1.828** [1.026, 3.257]	1.979** [1.103, 3.551]
Active nonfinancial-sanctions case, $t-1$	—	—	0.634 [0.267, 1.505]
E-government capacity, 10 pp	1.288* [0.974, 1.701]	1.285* [0.973, 1.697]	1.209 [0.900, 1.625]
Inflation	0.997 [0.975, 1.021]	0.996 [0.974, 1.020]	0.997 [0.974, 1.020]
Real GDP (constant 2015 USD), logged and standardized	1.120 [0.741, 1.693]	1.141 [0.752, 1.731]	1.188 [0.771, 1.832]
p: financial = nonfinancial	—	—	0.055
Jurisdiction–years	618	618	618
Jurisdictions	133	133	133
Commitments	64	64	64

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Hazard ratios; 95% confidence intervals in brackets.

5.1 Initiation of CBDC Investigation

Figure 2 presents the four informational-constraint measures from the preferred full initiation model. The upper panels show the structural measures, while the lower panels show the strategic measures. The left panels present domestic constraints, and the right panels present international constraints.

Domestic structural change is the clearest indicator of CBDC initiation. As shown in the upper-left panel, greater account ownership is associated with a higher hazard of initiating a CBDC investigation. A ten-percentage-point increase in account ownership is associated with an approximately 22 percent higher initiation hazard (HR = 1.22, $p < .001$). This result supports *H1*: jurisdictions with more extensive account-based finance tend to begin formal CBDC investigations earlier.

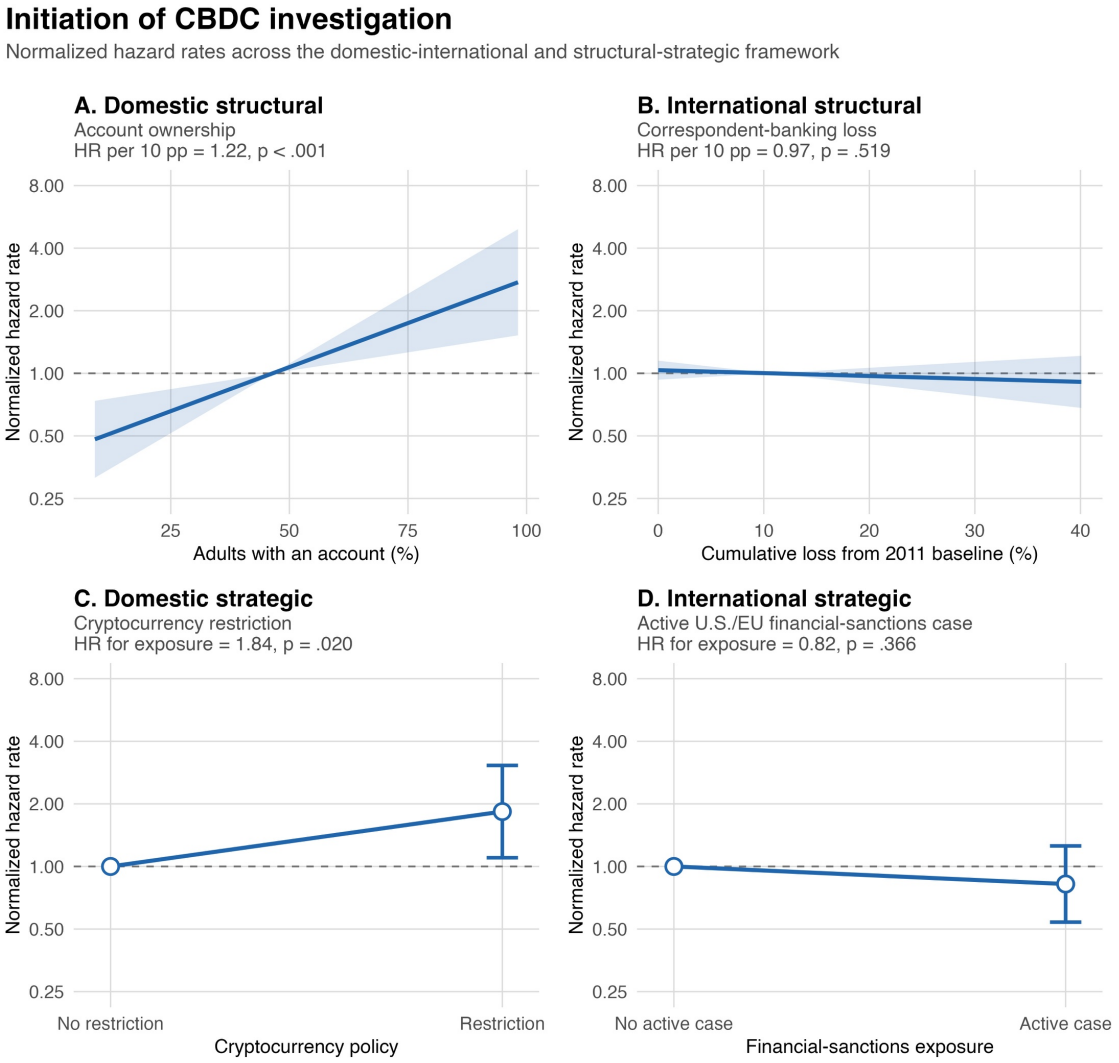


Figure 2: Informational constraints and the initiation of CBDC investigation

The upper-right panel shows little evidence of an association between correspondent-banking contraction and initiation (HR = 0.97, $p = .519$). The estimated association is close to the null and runs slightly in the opposite direction from that proposed in *H2a*. Prior FATF exposure is not displayed in the figure, but the preferred full model in Table 5 reports a negative association with initiation (HR = 0.811, 95% CI = $[0.653, 1.007]$, $p = .058$). The estimate is marginally significant at the ten-percent level but not at the five-percent level. Because its direction is negative, the result does not support the positive association proposed in *H2b*.

The lower-left panel shows stronger evidence of domestic strategic contestation. Jurisdictions with cryptocurrency restrictions have an estimated initiation hazard 84 percent higher than jurisdictions without such restrictions (HR = 1.84, $p = .020$). This full-model result supports *H3a*, which expects jurisdictions with cryptocurrency restrictions to initiate CBDC investigations earlier.

International strategic pressure has no comparable association with initiation. As shown in the lower-right panel, active financial-sanctions exposure is not significantly associated with initiation (HR = 0.82, $p = .366$). Because *H4* concerns commitment after an investigation has begun, this initiation estimate provides a baseline for comparison with the commitment results.

5.2 Commitment to a Pilot or Issuance

The pattern changes after a CBDC investigation has begun. Figure 3 presents the four informational-constraint measures from the preferred full commitment model. The upper panels show the structural measures, while the lower panels show the strategic measures. The left panels present domestic constraints, and the right panels present international constraints.

Neither structural measure is significantly associated with commitment. The upper-left panel shows a small and statistically insignificant association between account ownership and commitment (HR = 1.09, $p = .455$). The upper-right panel shows a similarly small and statistically insignificant association for correspondent-banking contraction (HR = 1.08, $p = .473$). The structural conditions associated with initiation do not have comparable associations with advancement toward a pilot or issuance.

The lower-left panel shows a positive but statistically insignificant association between cryptocurrency restrictions and commitment. Jurisdictions with restrictions have an estimated commitment hazard approximately twice that of jurisdictions without restrictions (HR = 2.04, $p = .169$). The wide confidence interval includes one. The complete estimates in Table 4 also vary across specifications. Cryptocurrency restrictions are positively and significantly associated with commitment in the Domestic specification (HR = 2.574, 95% CI = $[1.092, 6.063]$, $p = .031$), but not in the Strategic specification (HR = 1.862, 95% CI = $[0.783, 4.426]$, $p = .159$).

The variation appears to concern precision and magnitude rather than direction, since the estimated association remains positive in every specification. One possible explanation is the limited number of commitment events: the commitment risk set contains only 64 events, producing wide confidence intervals. In addition, cryptocurrency restrictions may overlap with financial-sanctions exposure or other forms of strategic pressure. Adding these measures in the Strategic and full specifications may separate associations that the Domestic

specification attributes to cryptocurrency restrictions alone. The restriction indicator also combines policies adopted for different reasons, which may further reduce the precision of the estimate. The evidence for *H3b* is therefore positive but specification-dependent.

The clearest commitment result appears in the lower-right panel. Among jurisdictions already investigating a CBDC, exposure to an active U.S. or EU financial-sanctions case in the preceding year is associated with an 82.8 percent higher hazard of advancing to a pilot or issuance (HR $= 1.828,$ 95% CI $= [1.026, 3.257],$ $p = .041$). This result supports *H4*, which expects active U.S. or EU financial sanctions to be associated with faster commitment among jurisdictions already investigating a CBDC.

Figure 4 examines whether this association is specific to financial sanctions. It compares the initiation and commitment estimates across the Broad, Financial, and Decomposed specifications. Blue circles represent initiation, while green squares represent commitment.

The upper-left panel shows that the broad indicator covering all active U.S. and EU sanctions cases is not associated with initiation (HR $= 0.92,$ $p = .683$). Its commitment estimate is positive but only marginally significant (HR $= 1.69,$ $p = .088$). The upper-right panel shows a clearer stage difference for financial sanctions. The comparison confirms the results found above: financial sanctions exposure is not associated with initiation (HR $= 0.82,$ $p = .366$), but it is positively associated with commitment (HR $= 1.83,$ $p = .041$).

The lower panels decompose sanctions exposure by type. The lower-left panel shows that active financial-sanctions cases are associated with a 97.9 percent higher commitment hazard (HR $= 1.979,$ 95% CI $= [1.103, 3.551],$ $p = .022$). The lower-right panel shows no comparable association for non-financial cases (HR $= 0.634,$ 95% CI $= [0.267, 1.505],$ $p = .301$). The difference between the financial and non-financial coefficients is suggestive ($p = .055$), although it narrowly misses the five-percent significance threshold. These results indicate that the commitment association is concentrated in sanctions affecting financial relations rather than sanctions exposure generally. This comparison provides the more specific test of *H4*.

The implementation capacity also points in the expected direction. In the preferred Financial specification, a 0.10-point increase in the E-Government Development Index is associated with a 28.5 percent higher commitment hazard (HR $= 1.285,$ 95% CI $= [0.973, 1.697],$ $p = .077$). The estimate is marginally significant at the ten-percent level but not at the five-percent level. Account ownership, real GDP, inflation, correspondent-banking contraction, and prior FATF exposure are not systematically associated with commitment. Because the commitment models contain only 64 events, the precision attainable for individual covariates is limited.

In sum, the results support *H1* and *H4*. They provide no support for *H2a* or *H2b*, while support for *H3a* and *H3b* depends on specification and timing. Domestic structural constraints are associated with CBDC investigation, whereas international strategic constraints are associated with advancement toward implementation.

Robustness. The Supplementary Appendix evaluates the two principal findings using alternative measures, timing rules, samples, estimators, and influence diagnostics. The account-ownership result remains positive across alternative Findex constructions, digital-payment use, bank-ownership interactions, and project types. The financial-sanctions estimate remains positive across the principal timing, geopolitical, estimator, and leave-one-jurisdiction-out checks, although its precision declines in some smaller or expanded samples.

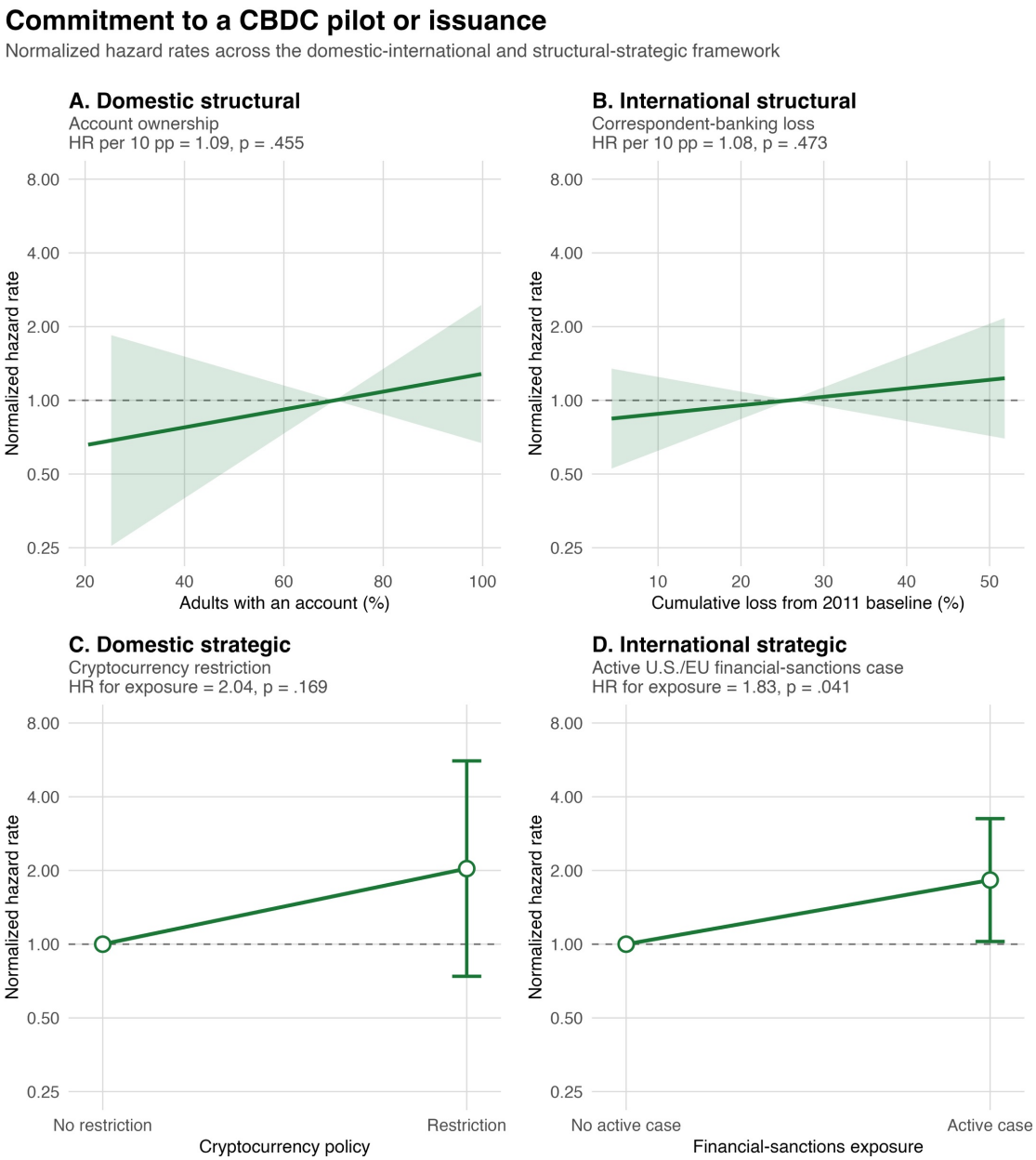


Figure 3: Informational constraints and commitment to a CBDC pilot or issuance

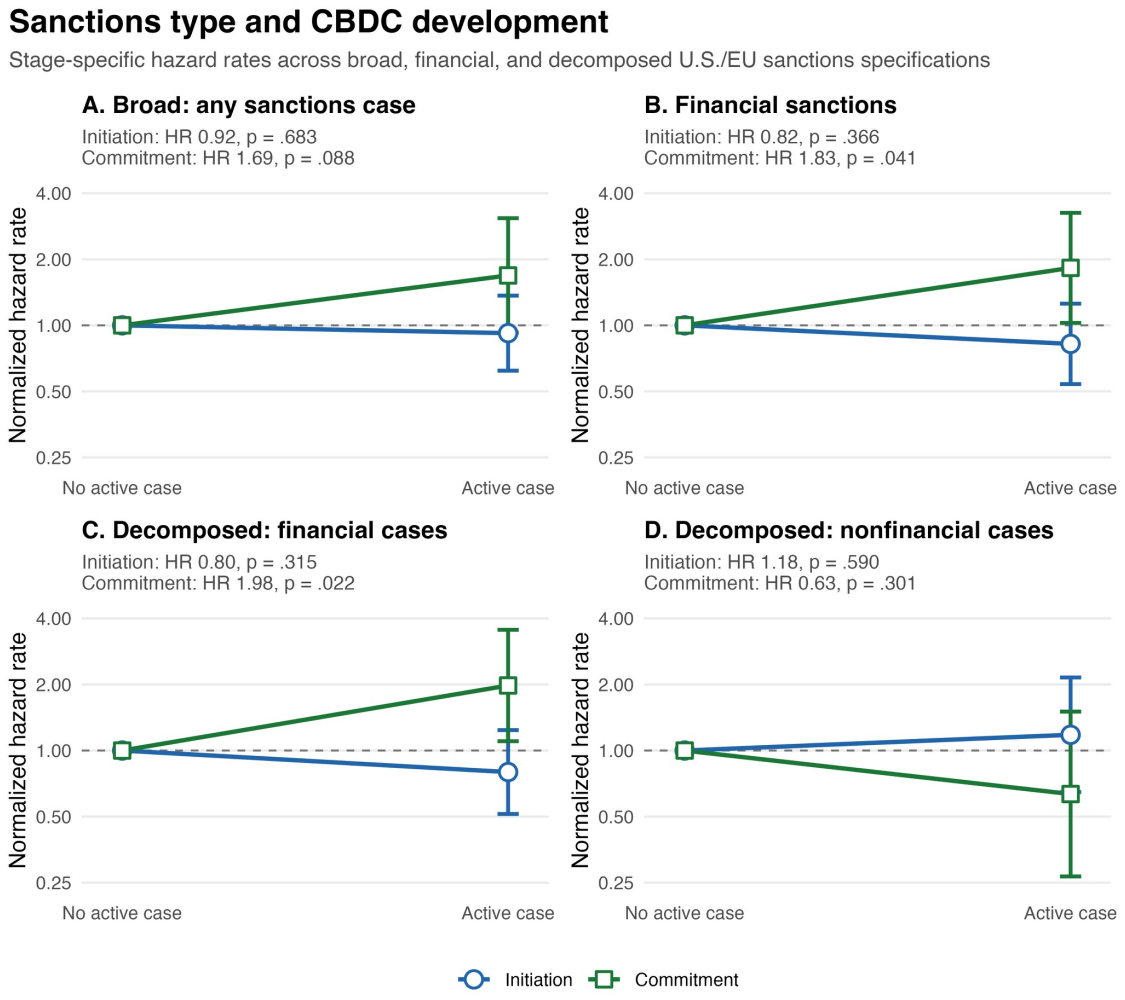


Figure 4: Stage-specific associations between U.S./EU sanctions and CBDC development

6 Discussion

6.1 Structural Change and Policy Learning

As account-based finance expands, payment records must be coordinated across a wider institutional network. Investigating a CBDC allows authorities to consider whether responsibilities for settlement, interoperability, identification, and data use should be reorganized. These constraints can motivate investigation without committing an authority to issuance.

This pattern is reflected in official reports from different monetary systems. Reports from Sweden, the euro area, and the United Kingdom highlight declining cash use, fragmented or outdated payment arrangements, and the migration of payments toward commercial-bank money and privately operated digital infrastructures[92]. These concerns are not limited to the distinction between banked and unbanked populations. In the Bahamas, roughly nine in ten adults already held an account by 2021[93]. Governor Rolle nonetheless situated the Sand Dollar, the Bahamian CBDC, within a broader modernization program addressing shrinking branch networks, geographic barriers in the Family Islands, and fragmented service provision across banks, credit unions, money-transfer businesses, and non-bank payment providers[94]. A 2025 follow-up connected it to a planned fast-payment platform spanning all retail payment providers and defined inclusion in terms of equitable access and cost rather than account possession[95]. High account ownership can provide the institutional base for CBDC development without eliminating informational, interoperability, and service-delivery constraints.

The same coordination problem becomes more pronounced as financial inclusion expands through multiple providers and technologies. Ghana's eCedi design paper connects the growth of mobile-money accounts and agents to interoperability and the supervision of bank and non-bank providers[96]. Nigeria's eNaira design builds on agent banking, standardized identification, a national payment switch, and fintech providers, while assigning customer-facing and compliance functions to regulated intermediaries[97]. In both cases, the problem is coordinating information across expanding networks. The major design concerns show the emphasis of governments on a state-legible digital payment network.

Kenya and the United States show that these demands need not produce commitment. Kenya concluded that its mobile-money ecosystem could address immediate payment problems without a CBDC, while identifying standards, APIs, customer identification, and cross-border interoperability as continuing concerns[98]. The Federal Reserve evaluated similar questions but required a CBDC to outperform available alternatives[99]. Project Hamilton, a technical research collaboration between the Federal Reserve Bank of Boston and MIT's Digital Currency Initiative, tested transaction-processing architectures, while FedNow offered a

[92]Sveriges Riksbank 2018; European Central Bank 2025a; Bank of England and HM Treasury 2024.
[93]Demirgüç-Kunt et al. 2022.
[94]Rolle 2021.
[95]Rolle 2025.
[96]Bank of Ghana 2022; Bank of Ghana 2024.
[97]Central Bank of Nigeria 2021.
[98]Central Bank of Kenya 2023.
[99]Board of Governors of the Federal Reserve System 2022.

competing route to modernizing interbank settlement[100]. Informational constraints can motivate investigation, while existing infrastructure and alternative reforms shape what follows.

6.2 Why Cross-Border Retrenchment Does Not Automatically Produce a CBDC

The null results for correspondent-banking contraction and FATF exposure are informative rather than disappointing. The measures identify cross-border constraints, and official records confirm that authorities recognized them. What they do not do is predict the choice of a CBDC. An authority facing correspondent withdrawal can seek new correspondents, strengthen compliance, connect payment systems, build a regional platform, or pursue a CBDC. The null estimates therefore distinguish between the existence of a constraint and the selection of a CBDC as the response.

Tonga illustrates this distinction. Its central bank linked bank de-risking to the need to preserve correspondent relationships, but its response combined AML/CFT supervision, a national KYC framework, regional cooperation, fintech development, domestic clearing modernization, and CBDC feasibility work as parallel elements of a broader program[101].

The wider Pacific experience echoes Tonga's case. Pacific island countries lost roughly 60 percent of their correspondent-banking relationships during the decade preceding 2024. Fiji, Kiribati, the Marshall Islands, Samoa, Tonga, Tuvalu, and Vanuatu responded not with immediate CBDC programs but with a temporary facility to preserve last-resort correspondent access, alongside AML/CFT reforms, regulatory harmonization, and regional aggregation[102]. This experience shows that severe network contraction can produce institutional reform without generating a CBDC in the short term.

The BIS unified-ledger blueprint would coordinate messaging, compliance, and settlement on one programmable venue, while noting that some gains are reachable more incrementally through API connections among existing systems[103]. IMF researchers propose a multilateral platform built on a common ledger and smart contracts[104]. Each adopts part of the underlying technology without constituting a CBDC programme, and each reorganizes the same informational functions differently. ASEAN makes the same point regionally: rather than a common CBDC, members have expanded bilateral QR and fast-payment links and are building Project Nexus to connect the instant-payment systems of India, Malaysia, the Philippines, Singapore, and Thailand[105].

The menu is wide, and in some cases losing correspondent relationships may itself weaken the resources needed to build a CBDC. This heterogeneity explains why a serious cross-border problem can be plainly visible in official documents without producing a uniform association

[100]Federal Reserve Bank of Boston 2022; Board of Governors of the Federal Reserve System 2023; U.S. Department of the Treasury 2022.

[101]National Reserve Bank of Tonga 2022.

[102]World Bank 2024.

[103]Bank for International Settlements 2023.

[104]Adrian et al. 2022.

[105]Bank for International Settlements 2024b; Association of Southeast Asian Nations 2023; Asian Development Bank 2023.

with either transition.

6.3 Strategic Pressure and the Conversion of Research into Commitment

The sanctions result shows why informational constraints matter strategically. Active U.S./EU financial-sanctions exposure accompanies faster commitment among jurisdictions that have already initiated an investigation. The U.S.-only estimate is positive and statistically significant, whereas the EU-only estimate is not. This pattern suggests that the association is concentrated in U.S.-sender cases, although the difference between the two estimates is not itself statistically established. Of the 133 jurisdictions in the complete-case commitment risk set, 71 experience an active financial-sanctions case in at least one risk-set year, and 30 of the 64 transitions to a pilot or issuance occur while the lagged indicator is active. These figures are consistent with the estimated association but do not establish causation.

Russia illustrates this sequence. The Bank of Russia opened public consultation on the digital ruble in October 2020 and developed a concept and prototype in 2021, before the major sanctions imposed in 2022[106]. After those sanctions restricted Russian banks' access to reserve-currency payment and settlement networks, Governor Nabiullina assigned the digital ruble a "special role" in domestic and prospective cross-border payments. She presented it alongside SPFS and bilateral national-currency arrangements as part of a broader set of responses[107]. The Bank's 2022 annual report placed the pilot within the development of domestically controlled, sanctions-resilient financial infrastructure[108]. Testing with real transactions began in August 2023 with thirteen banks[109]. This sequence illustrates the pattern behind the stage-specific estimates. Once a technically viable project exists, restrictions on payment and settlement networks can increase both its strategic value and the urgency of piloting it.

Alternative infrastructures do not provide an immediate or costless exit. Incumbent currencies and networks retain advantages of market depth, liquidity, institutional credibility, and scale. Investment in alternatives can nevertheless gradually expand a state's outside options and reduce the cost of later diversification. Continued geopolitical competition and uncertainty about U.S. financial statecraft may encourage authorities to commit more resources to CBDCs and other monetary arrangements.

6.4 Conclusion

CBDC development, this article has argued, is a staged response to constraints on informational reach within monetary systems. Digitalization does not merely increase the volume of monetary data. It reorganizes the institutional channels through which payment instructions and records are produced, transmitted, standardized, and used. Monetary authority depends not only on issuing currency or owning infrastructure but also on governing information flows across banks, platforms, payment providers, and jurisdictions. Informational constraints identify a dimension common to payment modernization: it changes how monetary information

[106]Bank of Russia 2020.
[107]Nabiullina 2022.
[108]Bank of Russia 2023a.
[109]Bank of Russia 2023b.

is distributed and thereby makes the distribution of institutional and political power subject to renegotiation.

The stage-specific findings show that informational constraints enter CBDC development in different ways. The expansion of account-based finance, a domestic structural issue, increases the value of investigation. Financial sanctions, an international strategic problem, increase the urgency of advancing a project that already exists. Investigation and commitment are not two degrees of the same decision but distinct forms of institutional adaptation. The former evaluates feasibility and develops options; the latter requires administrative capacity, legal authority, and coordination among public and private actors. The results show that informational constraints operate differently across structural and strategic domains, shaping both when authorities investigate alternative payment arrangements and when they advance existing projects toward implementation.

The implications extend more broadly to international political economy. Control over monetary information is increasingly embedded in technical standards, payment interfaces, identity systems, messaging networks, and settlement arrangements. Institutions positioned within these infrastructures shape not only whether transactions are visible but also whether payment instructions can be transmitted, authenticated, cleared, and settled. Digitalization changes the infrastructural sources through which monetary power is exercised. Dependence on foreign or privately governed infrastructures can create vulnerability, while the capacity to reorganize information flows can expand a state's strategic options. The politics of digital money is inseparable from wider contests over infrastructure, interdependence, and the governance of technology.

To my knowledge, this study is among the first to examine CBDC development through a systematic cross-national empirical framework. Its findings open several avenues for future research. One concerns what happens after commitment. Direct, hybrid, and intermediated architectures distribute responsibility for monetary informational governance differently among central banks, commercial institutions, and technology providers. Design choices not only involve privacy, offline functionality, interoperability, and programmability but also determine who can produce, transmit, access, and use monetary information—a political economy question closely tied to institutional change and the distribution of power.

A second avenue concerns the interaction between CBDCs and other forms of digital money. CBDCs should be examined alongside stablecoins, cryptocurrencies, instant-payment systems, and multilateral settlement platforms. Each offers a different arrangement for organizing monetary information. Their interaction may reshape political coalitions, technical standards, and the hierarchy of cross-border payment networks. Digital monetary politics concerns not only who issues money but also who governs the information and channels that make money operational. CBDCs are a particularly consequential state response. Their uneven development demonstrates that this is a political rather than merely technical process.

The global CBDC wave is more than a technological modernization of money. It reflects a political contest over who controls the information embedded in monetary systems and, by extension, who can exercise monetary power in the digital economy. As money becomes increasingly data-driven and AI expands the possibilities and blurs the boundaries of data use, control over monetary information will become an increasingly important source of power alongside control over credit, infrastructure, and coercive capacity.

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